

BENCHMARK: MARGINAL

Results

Mid-Transit Time (Tmid)	2459969.7745 +/- 0.0029 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.1667 +/- 0.01
Transit depth (Rp/Rs)^2	2.78 +/- 0.33 [%]
Orbital Inclination (inc)	81.68 +/- 0.54
Airmass coefficient 1 (a1)	1.154 +/- 0.017
Airmass coefficient 2 (a2)	-0.1011 +/- 0.0093
Scatter in the residuals of the lightcurve fit is	1.44 %
Best Comparison Star	None
Optimal Aperture	4.27
Optimal Annulus	5.56
Transit Duration (day)	0.0676 +/- 0.0051

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.1667 ± 0.01 vs 0.1368 (deviation 2.72σ)
Duration fit vs published	0.0676 d vs 0.0758 d (deviation 1.47σ)
Mid-time O–C	-5.4 min
Depth SNR	15.2
Residual scatter	1.44 %

Light curve

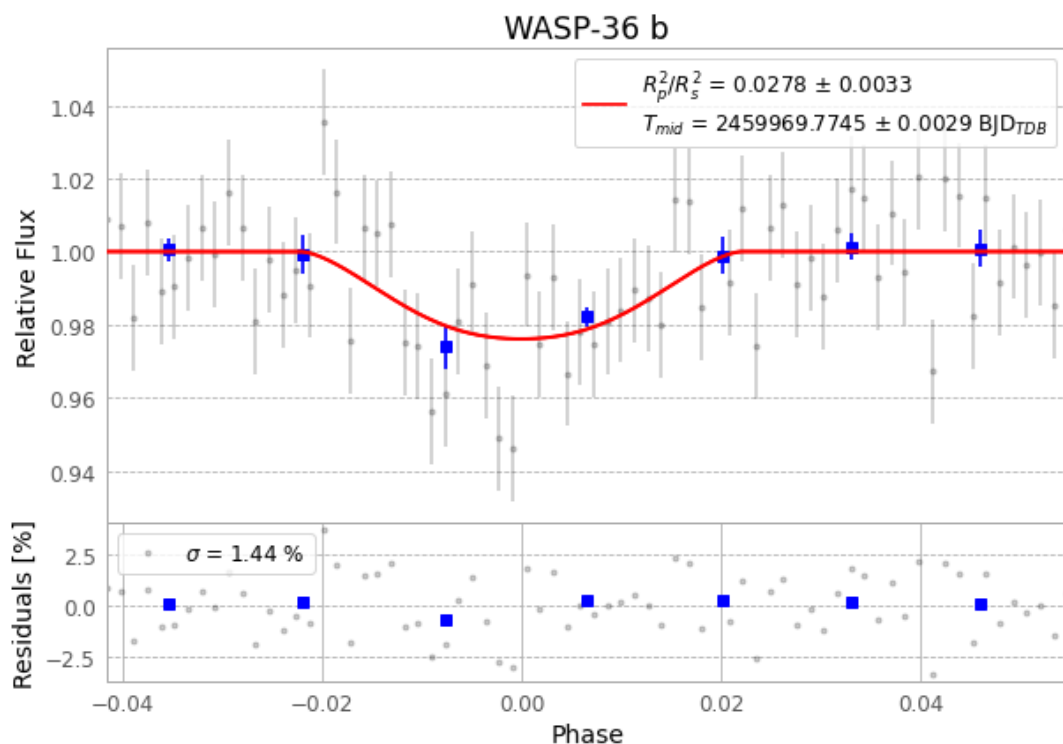


Figure 1 — FinalLightCurve_WASP-36 b_2023-01-24.png (SHA-256 51e8d4c0b3222309...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [371, 83], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 126d71ed6d706abb...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ead60b9

Data provenance

72 FITS frames (47 MB), WASP-36230125045415.FITS → WASP-36230125082715.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-WASP-36-230125.log (4f3cc339cbf9...), FinalLightCurve_WASP-36 b_2023-01-24.png (51e8d4c0b322...), FinalLightCurve_WASP-36 b_2023-01-24.csv (e110b762d37a...)

Compute resources

Wall time 180.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-07-28 21:02Z.